

Summer Bootcamp

Session 2 – Calculus and ODE

Differentiation, integration, Taylor expansion, and financial dynamics

Nathan Sauldubois

NYU Finance and Risk Engineering

August 2026

Learning goals

By the end of the session, students should be able to:

- compare approximation errors using Landau notation;
- compute derivatives, gradients, Jacobians, and Hessians;
- use Taylor expansion for approximation and local analysis;
- manipulate integrals, substitutions, and differentiation under the integral sign;
- solve scalar and multidimensional linear ODEs;
- connect ODEs to risk-free growth, discounting, and Vasicek mean reversion.

Section Roadmap

1 Landau Notation and Approximation

Landau Notation Compares Scales

Let f and g be defined near a , with $g(x) \neq 0$ for $x \neq a$.

Big O : bounded relative size

We write $f(x) = O(g(x))$ as $x \rightarrow a$ if there exist $C, \delta > 0$ such that

$$0 < |x - a| < \delta \implies |f(x)| \leq C|g(x)|.$$

Landau Notation Compares Scales

Let f and g be defined near a , with $g(x) \neq 0$ for $x \neq a$.

Big O : bounded relative size

We write $f(x) = O(g(x))$ as $x \rightarrow a$ if there exist $C, \delta > 0$ such that

$$0 < |x - a| < \delta \implies |f(x)| \leq C|g(x)|.$$

Small o : negligible relative size

We write $f(x) = o(g(x))$ as $x \rightarrow a$ if

$$\frac{f(x)}{g(x)} \longrightarrow 0.$$

Thus $O(g)$ means “at most the scale of g ”, whereas $o(g)$ means “strictly smaller than the scale of g ”.

Examples in One Dimension

As $h \rightarrow 0$, determine the correct O or o relation for each expression:

$$h^3 \text{ versus } h^2, \quad 3h^2 - h^3 \text{ versus } h^2, \quad \sin h - h \text{ versus } h^2.$$

Examples in One Dimension

As $h \rightarrow 0$, determine the correct O or o relation for each expression:

$$h^3 \text{ versus } h^2, \quad 3h^2 - h^3 \text{ versus } h^2, \quad \sin h - h \text{ versus } h^2.$$

The answers are

$$h^3 = o(h^2), \quad 3h^2 - h^3 = O(h^2), \quad \sin h = h + O(h^3).$$

Examples in One Dimension

As $h \rightarrow 0$, determine the correct O or o relation for each expression:

$$h^3 \text{ versus } h^2, \quad 3h^2 - h^3 \text{ versus } h^2, \quad \sin h - h \text{ versus } h^2.$$

The answers are

$$h^3 = o(h^2), \quad 3h^2 - h^3 = O(h^2), \quad \sin h = h + O(h^3).$$

Indeed,

$$\frac{h^3}{h^2} = h \rightarrow 0, \quad \left| \frac{3h^2 - h^3}{h^2} \right| = |3 - h| \leq 4$$

for $|h| < 1$.

Examples in One Dimension

As $h \rightarrow 0$, determine the correct O or o relation for each expression:

$$h^3 \text{ versus } h^2, \quad 3h^2 - h^3 \text{ versus } h^2, \quad \sin h - h \text{ versus } h^2.$$

The answers are

$$h^3 = o(h^2), \quad 3h^2 - h^3 = O(h^2), \quad \sin h = h + O(h^3).$$

Indeed,

$$\frac{h^3}{h^2} = h \rightarrow 0, \quad \left| \frac{3h^2 - h^3}{h^2} \right| = |3 - h| \leq 4$$

for $|h| < 1$.

The distinction matters

$h^2 = O(h^2)$, but $h^2 \neq o(h^2)$ because the ratio equals 1. Moreover, $o(g) \subset O(g)$ near the limit point.

Landau Notation Encodes Local Expansions

As $h \rightarrow 0$,

$$e^h = 1 + h + \frac{h^2}{2} + O(h^3), \quad \log(1 + h) = h - \frac{h^2}{2} + o(h^2).$$

Landau Notation Encodes Local Expansions

As $h \rightarrow 0$,

$$e^h = 1 + h + \frac{h^2}{2} + O(h^3), \quad \log(1 + h) = h - \frac{h^2}{2} + o(h^2).$$

For a differentiable function at a ,

$$f(a + h) = f(a) + f'(a)h + o(|h|).$$

The remainder is negligible compared with the linear term's natural scale.

Landau Notation Encodes Local Expansions

As $h \rightarrow 0$,

$$e^h = 1 + h + \frac{h^2}{2} + O(h^3), \quad \log(1 + h) = h - \frac{h^2}{2} + o(h^2).$$

For a differentiable function at a ,

$$f(a + h) = f(a) + f'(a)h + o(|h|).$$

The remainder is negligible compared with the linear term's natural scale.

Example

Use $\sqrt{1 + h} = 1 + \frac{1}{2}h + O(h^2)$ to expand $\sqrt{4 + x}$ as $x \rightarrow 0$.

Landau Notation Encodes Local Expansions

As $h \rightarrow 0$,

$$e^h = 1 + h + \frac{h^2}{2} + O(h^3), \quad \log(1 + h) = h - \frac{h^2}{2} + o(h^2).$$

For a differentiable function at a ,

$$f(a + h) = f(a) + f'(a)h + o(|h|).$$

The remainder is negligible compared with the linear term's natural scale.

Example

Use $\sqrt{1 + h} = 1 + \frac{1}{2}h + O(h^2)$ to expand $\sqrt{4 + x}$ as $x \rightarrow 0$.

$$\sqrt{4 + x} = 2\sqrt{1 + \frac{x}{4}} = 2 + \frac{x}{4} + O(x^2) \quad (x \rightarrow 0).$$

The Same Definitions Work in $\mathbb{R}^d$

Let $F : \mathbb{R}^d \rightarrow \mathbb{R}^m$ and let $G : \mathbb{R}^d \rightarrow [0, \infty)$. Using any norm, as $h \rightarrow 0$ in $\mathbb{R}^d$,

$$F(h) = O(G(h)) \iff \|F(h)\| \leq CG(h) \quad \text{near } 0,$$

$$F(h) = o(G(h)) \iff \frac{\|F(h)\|}{G(h)} \longrightarrow 0.$$

The Same Definitions Work in $\mathbb{R}^d$

Let $F : \mathbb{R}^d \rightarrow \mathbb{R}^m$ and let $G : \mathbb{R}^d \rightarrow [0, \infty)$. Using any norm, as $h \rightarrow 0$ in $\mathbb{R}^d$,

$$F(h) = O(G(h)) \iff \|F(h)\| \leq CG(h) \quad \text{near } 0,$$

$$F(h) = o(G(h)) \iff \frac{\|F(h)\|}{G(h)} \longrightarrow 0.$$

Polynomial examples

For $h = (h_1, h_2) \rightarrow 0$, compare $h_1 h_2$ and $h_1^2 h_2$ with $\|h\|^2$.

The Same Definitions Work in $\mathbb{R}^d$

Let $F : \mathbb{R}^d \rightarrow \mathbb{R}^m$ and let $G : \mathbb{R}^d \rightarrow [0, \infty)$. Using any norm, as $h \rightarrow 0$ in $\mathbb{R}^d$,

$$F(h) = O(G(h)) \iff \|F(h)\| \leq CG(h) \quad \text{near } 0,$$

$$F(h) = o(G(h)) \iff \frac{\|F(h)\|}{G(h)} \longrightarrow 0.$$

Polynomial examples

For $h = (h_1, h_2) \rightarrow 0$, compare $h_1 h_2$ and $h_1^2 h_2$ with $\|h\|^2$.

$$h_1 h_2 = O(\|h\|^2), \quad h_1^2 h_2 = o(\|h\|^2).$$

Indeed, $|h_1 h_2| \leq \frac{1}{2} \|h\|^2$ and $|h_1^2 h_2| \leq \|h\|^3 = o(\|h\|^2)$.

Section Roadmap

2 One-Dimensional Differentiation

Derivative: Definition and Meaning

Definition

For a function $f : \mathbb{R} \rightarrow \mathbb{R}$,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h},$$

when the limit exists.

Derivative: Definition and Meaning

Definition

For a function $f : \mathbb{R} \rightarrow \mathbb{R}$,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h},$$

when the limit exists.

- Instantaneous rate of change.
- Slope of the tangent line at $(x, f(x))$.
- Local linear approximation:

$$f(x+h) \approx f(x) + f'(x)h.$$

C^1 Regularity in One Dimension

Let $I \subset \mathbb{R}$ be an open interval and let $f : I \rightarrow \mathbb{R}$.

Definition

$f \in C^1(I)$ if f is differentiable at every point of I and $f' : I \rightarrow \mathbb{R}$ is continuous.

C^1 Regularity in One Dimension

Let $I \subset \mathbb{R}$ be an open interval and let $f : I \rightarrow \mathbb{R}$.

Definition

$f \in C^1(I)$ if f is differentiable at every point of I and $f' : I \rightarrow \mathbb{R}$ is continuous.

Differentiability alone does not imply C^1 regularity: continuity of the derivative is an additional requirement.

Quick check

$x \mapsto e^x$ and $x \mapsto |x|^3$ are in $C^1(\mathbb{R})$, whereas $x \mapsto |x|$ is not differentiable at 0.

Differentiation Rules

For differentiable functions f, g and constants a, b :

$$(af + bg)' = af' + bg',$$

$$(fg)' = f'g + fg', \quad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.$$

Differentiation Rules

For differentiable functions f, g and constants a, b :

$$(af + bg)' = af' + bg',$$

$$(fg)' = f'g + fg', \quad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.$$

Result – Chain rule

If $F(x) = f(g(x))$, then

$$F'(x) = f'(g(x))g'(x).$$

Differentiation Rules

For differentiable functions f, g and constants a, b :

$$(af + bg)' = af' + bg',$$

$$(fg)' = f'g + fg', \quad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.$$

Result – Chain rule

If $F(x) = f(g(x))$, then

$$F'(x) = f'(g(x))g'(x).$$

Example

Compute the derivative of $x \mapsto (2x + 1)^4$.

Differentiation Rules

For differentiable functions f, g and constants a, b :

$$(af + bg)' = af' + bg',$$

$$(fg)' = f'g + fg', \quad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.$$

Result – Chain rule

If $F(x) = f(g(x))$, then

$$F'(x) = f'(g(x))g'(x).$$

Example

Compute the derivative of $x \mapsto (2x + 1)^4$.

$$\frac{d}{dx}(2x + 1)^4 = 4(2x + 1)^3 \cdot 2 = 8(2x + 1)^3.$$

Classic Derivatives

Function	Derivative
x^n	nx^{n-1}
e^x	e^x
$\log x$	$1/x$, for $x > 0$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$

One-Dimensional Derivative Toolkit

For differentiable f, g and constant a :

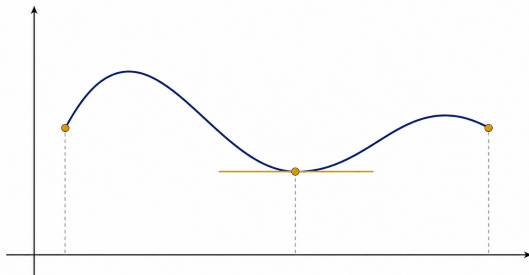
$$\begin{aligned}(fg)' &= f'g + fg', & \left(\frac{f}{g}\right)' &= \frac{f'g - fg'}{g^2}, \\(f \circ g)' &= (f' \circ g)g', & (g^a)' &= ag^{a-1}g', \\(e^g)' &= e^g g', & (\log g)' &= \frac{g'}{g}.\end{aligned}$$

Rolle's Theorem

Result

If f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then there exists $c \in (a, b)$ such that

$$f'(c) = 0.$$



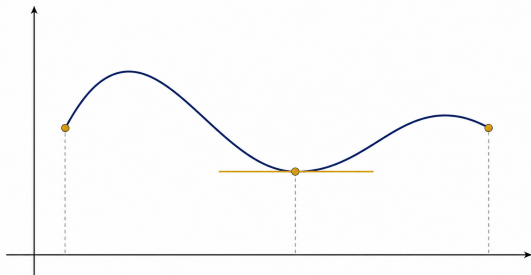
$$f(a) = f(b), \quad f'(c) = 0.$$

Rolle's Theorem

Result

If f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then there exists $c \in (a, b)$ such that

$$f'(c) = 0.$$



$$f(a) = f(b), \quad f'(c) = 0.$$

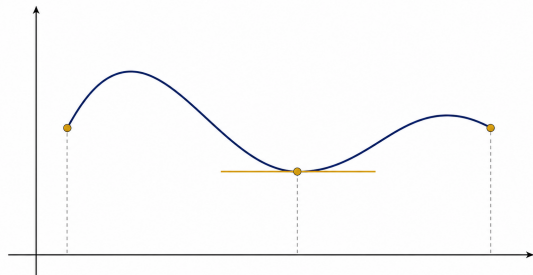
For $f(x) = x^2 - 4x$ on $[0, 4]$, find a point guaranteed by Rolle's theorem.

Rolle's Theorem

Result

If f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then there exists $c \in (a, b)$ such that

$$f'(c) = 0.$$



$$f(a) = f(b), \quad f'(c) = 0.$$

For $f(x) = x^2 - 4x$ on $[0, 4]$, find a point guaranteed by Rolle's theorem. Since $f'(x) = 2x - 4$, Rolle's point is $c = 2$.

Mean Value Theorem

Result – Mean value theorem

If f is continuous on $[a, b]$ and differentiable on (a, b) , then for some $c \in (a, b)$,

$$f(b) - f(a) = f'(c)(b - a).$$

The Sign of the Derivative Controls Monotonicity

Consequence – Monotonicity criteria

Let f be differentiable on an interval I .

$$\begin{aligned} f'(x) &\geq 0 \text{ on } I &\implies f \text{ is nondecreasing,} \\ f'(x) &> 0 \text{ on } I &\implies f \text{ is strictly increasing,} \\ f'(x) &\leq 0 \text{ on } I &\implies f \text{ is nonincreasing,} \\ f'(x) &= 0 \text{ on } I &\implies f \text{ is constant.} \end{aligned}$$

Indeed, for $x < y$, the mean value theorem gives a point $c \in (x, y)$ such that

$$f(y) - f(x) = f'(c)(y - x).$$

Since $y - x > 0$, the sign of the increment is determined by the sign of $f'(c)$.

A Derivative Bound Controls All Increments

Consequence – Lipschitz estimate

If $|f'(x)| \leq L$ on an interval I , then for every $x, y \in I$,

$$|f(y) - f(x)| \leq L|y - x|.$$

Thus f is L -Lipschitz continuous on I .

A Derivative Bound Controls All Increments

Consequence – Lipschitz estimate

If $|f'(x)| \leq L$ on an interval I , then for every $x, y \in I$,

$$|f(y) - f(x)| \leq L|y - x|.$$

Thus f is L -Lipschitz continuous on I .

Consequence – Uniqueness from strict monotonicity

If $f'(x) > 0$ everywhere on I , or $f'(x) < 0$ everywhere on I , then f is one-to-one. In particular,

$$f(x) = 0 \text{ has at most one solution in } I.$$

These conclusions are global even though the derivative is local information.

3 Multivariate Calculus

Partial Derivatives and Gradient

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the partial derivative with respect to x_i is

$$\partial_{x_i} f(x) = \lim_{h \rightarrow 0} \frac{f(x_1, \dots, x_i + h, \dots, x_n) - f(x)}{h}.$$

Partial Derivatives and Gradient

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the partial derivative with respect to x_i is

$$\partial_{x_i} f(x) = \lim_{h \rightarrow 0} \frac{f(x_1, \dots, x_i + h, \dots, x_n) - f(x)}{h}.$$

The gradient is

$$\nabla f(x) = \begin{pmatrix} \partial_{x_1} f(x) \\ \vdots \\ \partial_{x_n} f(x) \end{pmatrix}.$$

Partial Derivatives and Gradient

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the partial derivative with respect to x_i is

$$\partial_{x_i} f(x) = \lim_{h \rightarrow 0} \frac{f(x_1, \dots, x_i + h, \dots, x_n) - f(x)}{h}.$$

The gradient is

$$\nabla f(x) = \begin{pmatrix} \partial_{x_1} f(x) \\ \vdots \\ \partial_{x_n} f(x) \end{pmatrix}.$$

Directional derivative

For $\|u\| = 1$,

$$D_u f(x) = \nabla f(x)^\top u.$$

The gradient points toward steepest ascent.

The Differential Is a Linear Map

We say that $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable at x if there exists a *linear map*

$$Dg(x) : \mathbb{R}^n \longrightarrow \mathbb{R}^m.$$

such that

$$g(x + h) = g(x) + Dg(x)h + o(\|h\|).$$

The Differential Is a Linear Map

We say that $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable at x if there exists a *linear map*

$$Dg(x) : \mathbb{R}^n \longrightarrow \mathbb{R}^m.$$

such that

$$g(x + h) = g(x) + Dg(x)h + o(\|h\|).$$

Linearity means that, for increments h, k and scalars α, β ,

$$Dg(x)(\alpha h + \beta k) = \alpha Dg(x)h + \beta Dg(x)k.$$

Do not confuse the two maps

g may be nonlinear. Its differential $h \mapsto Dg(x)h$ is linear in the increment h once the base point x is fixed.

Differentiability in Several Variables

The linear map $Dg(x)$ is the best first-order approximation of g at x :

$$g(x + h) = g(x) + Dg(x)h + r(h), \quad \frac{\|r(h)\|}{\|h\|} \rightarrow 0.$$

Differentiability in Several Variables

The linear map $Dg(x)$ is the best first-order approximation of g at x :

$$g(x+h) = g(x) + Dg(x)h + r(h), \quad \frac{\|r(h)\|}{\|h\|} \rightarrow 0.$$

In coordinates, the matrix of $Dg(x)$ is the Jacobian $J_g(x)$:

$$Dg(x)h = J_g(x)h.$$

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, this becomes

$$f(x+h) = f(x) + \nabla f(x)^\top h + o(\|h\|).$$

Sufficient condition

Continuous first-order partial derivatives near x imply differentiability.

C^1 Regularity in Several Dimensions

Let $U \subset \mathbb{R}^n$ be open and let $g : U \rightarrow \mathbb{R}^m$.

Definition

$g \in C^1(U; \mathbb{R}^m)$ if g is differentiable on U and

$$x \mapsto Dg(x) \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$$

is continuous.

C^1 Regularity in Several Dimensions

Let $U \subset \mathbb{R}^n$ be open and let $g : U \rightarrow \mathbb{R}^m$.

Definition

$g \in C^1(U; \mathbb{R}^m)$ if g is differentiable on U and

$$x \mapsto Dg(x) \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$$

is continuous.

Equivalently, all first-order partial derivatives exist and are continuous:

$$\partial_{x_j} g_i \in C^0(U), \quad 1 \leq i \leq m, \quad 1 \leq j \leq n.$$

Thus $g \in C^1$ exactly when its Jacobian $J_g(x)$ varies continuously with x .

Exercise 1: Compute a Gradient with Partial Derivatives

For

$$f(x, y) = x^2 + 3xy + 2y^2, \quad \nabla f(x, y) = \begin{pmatrix} \\ \end{pmatrix}.$$

Exercise 1: Compute a Gradient with Partial Derivatives

For

$$f(x, y) = x^2 + 3xy + 2y^2, \quad \nabla f(x, y) = \begin{pmatrix} \\ \end{pmatrix}.$$

Compute one coordinate at a time:

$$\partial_x f(x, y) = 2x + 3y, \quad \partial_y f(x, y) = 3x + 4y.$$

Therefore

$$\boxed{\nabla f(x, y) = \begin{pmatrix} 2x + 3y \\ 3x + 4y \end{pmatrix}.$$

Vector-Valued Functions and Jacobian

For a map $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$,

$$g(x) = \begin{pmatrix} g_1(x) \\ \vdots \\ g_m(x) \end{pmatrix},$$

the Jacobian matrix is

$$J_g(x) = \begin{pmatrix} \partial_{x_1} g_1(x) & \cdots & \partial_{x_n} g_1(x) \\ \vdots & \ddots & \vdots \\ \partial_{x_1} g_m(x) & \cdots & \partial_{x_n} g_m(x) \end{pmatrix}.$$

Vector-Valued Functions and Jacobian

For a map $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$,

$$g(x) = \begin{pmatrix} g_1(x) \\ \vdots \\ g_m(x) \end{pmatrix},$$

the Jacobian matrix is

$$J_g(x) = \begin{pmatrix} \partial_{x_1} g_1(x) & \cdots & \partial_{x_n} g_1(x) \\ \vdots & \ddots & \vdots \\ \partial_{x_1} g_m(x) & \cdots & \partial_{x_n} g_m(x) \end{pmatrix}.$$

Interpretation

$J_g(x)$ is the best linear approximation of g near x .

Jacobian Example

Let

$$\Phi(u, v) = (ue^v, u^2 + v).$$

Compute

$$J_{\Phi}(u, v) = \underline{\hspace{2cm}}, \quad \det J_{\Phi}(u, v) = \underline{\hspace{2cm}}.$$

Jacobian Example

Let

$$\Phi(u, v) = (ue^v, u^2 + v).$$

Compute

$$J_{\Phi}(u, v) = \underline{\hspace{2cm}}, \quad \det J_{\Phi}(u, v) = \underline{\hspace{2cm}}.$$

Then

$$J_{\Phi}(u, v) = \begin{pmatrix} \partial_u(ue^v) & \partial_v(ue^v) \\ \partial_u(u^2 + v) & \partial_v(u^2 + v) \end{pmatrix} = \begin{pmatrix} e^v & ue^v \\ 2u & 1 \end{pmatrix}.$$

Jacobian Example

Let

$$\Phi(u, v) = (ue^v, u^2 + v).$$

Compute

$$J_{\Phi}(u, v) = \underline{\hspace{2cm}}, \quad \det J_{\Phi}(u, v) = \underline{\hspace{2cm}}.$$

Then

$$J_{\Phi}(u, v) = \begin{pmatrix} \partial_u(ue^v) & \partial_v(ue^v) \\ \partial_u(u^2 + v) & \partial_v(u^2 + v) \end{pmatrix} = \begin{pmatrix} e^v & ue^v \\ 2u & 1 \end{pmatrix}.$$

The determinant is

$$\det J_{\Phi}(u, v) = e^v(1 - 2u^2).$$

Exercise 2: Derive the Multivariate Chain Rule

Let $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $f : \mathbb{R}^m \rightarrow \mathbb{R}$.

Task

Find the first-order expansion of

$$(f \circ g)(x + h) - (f \circ g)(x)$$

and deduce $\nabla(f \circ g)(x)$.

Exercise 2: Derive the Multivariate Chain Rule

Let $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $f : \mathbb{R}^m \rightarrow \mathbb{R}$.

Task

Find the first-order expansion of

$$(f \circ g)(x + h) - (f \circ g)(x)$$

and deduce $\nabla(f \circ g)(x)$.

Step 1. First expand the inner function:

$$g(x + h) = g(x) + J_g(x)h + o(\|h\|).$$

Exercise 2: Derive the Multivariate Chain Rule

Let $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $f : \mathbb{R}^m \rightarrow \mathbb{R}$.

Task

Find the first-order expansion of

$$(f \circ g)(x + h) - (f \circ g)(x)$$

and deduce $\nabla(f \circ g)(x)$.

Step 1. First expand the inner function:

$$g(x + h) = g(x) + J_g(x)h + o(\|h\|).$$

Step 2. Set $k = J_g(x)h + o(\|h\|)$ and expand the outer function:

$$f(g(x) + k) = f(g(x)) + \nabla f(g(x))^T k + o(\|k\|).$$

Exercise 2: Derive the Multivariate Chain Rule

Let $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $f : \mathbb{R}^m \rightarrow \mathbb{R}$.

Task

Find the first-order expansion of

$$(f \circ g)(x + h) - (f \circ g)(x)$$

and deduce $\nabla(f \circ g)(x)$.

Step 1. First expand the inner function:

$$g(x + h) = g(x) + J_g(x)h + o(\|h\|).$$

Step 2. Set $k = J_g(x)h + o(\|h\|)$ and expand the outer function:

$$f(g(x) + k) = f(g(x)) + \nabla f(g(x))^T k + o(\|k\|).$$

Combining the two expansions gives

$$f(g(x + h)) - f(g(x)) = \nabla f(g(x))^T J_g(x)h + o(\|h\|).$$

Therefore

$$\nabla(f \circ g)(x) = J_g(x)^T \nabla f(g(x)).$$

The Same Chain Rule in Partial Derivatives

Write $g = (g_1, \dots, g_m)$. For every input coordinate x_j ,

$$\partial_{x_j}(f \circ g)(x) = \sum_{i=1}^m \partial_{y_i} f(g(x)) \partial_{x_j} g_i(x).$$

The Same Chain Rule in Partial Derivatives

Write $g = (g_1, \dots, g_m)$. For every input coordinate x_j ,

$$\partial_{x_j}(f \circ g)(x) = \sum_{i=1}^m \partial_{y_i} f(g(x)) \partial_{x_j} g_i(x).$$

This is exactly the j -th component of

$$J_g(x)^\top \nabla f(g(x)).$$

Thus the increment calculation and the coordinate calculation give the same formula: the Jacobian transports the first-order variation through the composition.

Scalar Product: Two Ways to Find the Differential

Let $p(x, y) = x^\top y$ for $x, y \in \mathbb{R}^n$.

Scalar Product: Two Ways to Find the Differential

Let $p(x, y) = x^\top y$ for $x, y \in \mathbb{R}^n$. Coordinatewise,

$$\partial_{x_i} p = y_i, \quad \partial_{y_i} p = x_i,$$

$$\nabla p(x, y) = \begin{pmatrix} y \\ x \end{pmatrix}.$$

Scalar Product: Two Ways to Find the Differential

Let $p(x, y) = x^\top y$ for $x, y \in \mathbb{R}^n$. Coordinatewise,

$$\partial_{x_i} p = y_i, \quad \partial_{y_i} p = x_i,$$

$$\nabla p(x, y) = \begin{pmatrix} y \\ x \end{pmatrix}.$$

For increments (h, k) ,

$$p(x + h, y + k) - p(x, y) = y^\top h + x^\top k + h^\top k.$$

The linear part is therefore

$$Dp(x, y)(h, k) = y^\top h + x^\top k.$$

Quadratic Form: Two Ways to Find the Gradient

Let $q(x) = x^\top Ax$, where $A \in \mathbb{R}^{n \times n}$.

Quadratic Form: Two Ways to Find the Gradient

Let $q(x) = x^\top Ax$, where $A \in \mathbb{R}^{n \times n}$. Using partial derivatives,

$$\partial_{x_i} q(x) = \sum_j A_{ij} x_j + \sum_j A_{ji} x_j, \quad \boxed{\nabla q(x) = (A + A^\top)x.}$$

Quadratic Form: Two Ways to Find the Gradient

Let $q(x) = x^\top Ax$, where $A \in \mathbb{R}^{n \times n}$. Using partial derivatives,

$$\partial_{x_i} q(x) = \sum_j A_{ij} x_j + \sum_j A_{ji} x_j, \quad \boxed{\nabla q(x) = (A + A^\top)x.}$$

Using an increment h ,

$$q(x + h) - q(x) = x^\top Ah + h^\top Ax + h^\top Ah.$$

Hence

$$Dq(x)h = ((A + A^\top)x)^\top h.$$

If A is symmetric, $\nabla q(x) = 2Ax$.

4 Implicit and Inverse Function Theorems

Result – Implicit function theorem

Suppose $F(x_0, y_0) = 0$ and

$$\partial_y F(x_0, y_0) \neq 0.$$

If F is continuously differentiable, then locally the equation $F(x, y) = 0$ defines a unique differentiable function $y = \varphi(x)$.

Implicit Function Theorem

Result – Implicit function theorem

Suppose $F(x_0, y_0) = 0$ and

$$\partial_y F(x_0, y_0) \neq 0.$$

If F is continuously differentiable, then locally the equation $F(x, y) = 0$ defines a unique differentiable function $y = \varphi(x)$. Differentiating $F(x, \varphi(x)) = 0$ gives

$$\boxed{\varphi'(x) = -\frac{\partial_x F(x, \varphi(x))}{\partial_y F(x, \varphi(x))}}.$$

Implicit Differentiation on the Unit Circle

For $F(x, y) = x^2 + y^2 - 1$, find dy/dx wherever the curve can be written locally as $y = \varphi(x)$.

Implicit Differentiation on the Unit Circle

For $F(x, y) = x^2 + y^2 - 1$, find dy/dx wherever the curve can be written locally as $y = \varphi(x)$.
Whenever $y \neq 0$,

$$\frac{dy}{dx} = -\frac{2x}{2y} = -\frac{x}{y}.$$

At $(1, 0)$ the condition fails: the tangent is vertical.

The Inverse Problem Is to Solve $g(x) = y$

Given a target $y \in \mathbb{R}^n$, we seek x such that

$$g(x) = y.$$

If $g : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is continuously differentiable and $J_g(x_0)$ is invertible, then nearby targets have a unique local solution $x = g^{-1}(y)$, with

$$J_{g^{-1}}(g(x_0)) = J_g(x_0)^{-1}.$$

Geometric meaning

An invertible Jacobian means that the local linear approximation neither flattens nor loses a direction.

Newton Solves the Linearized Equation

Set $F(x) = g(x) - y$. Starting from x_k , linearize:

$$F(x_k + s) \approx F(x_k) + J_g(x_k)s.$$

Impose $F(x_k) + J_g(x_k)s_k = 0$:

$$\boxed{J_g(x_k)s_k = y - g(x_k)}, \quad \boxed{x_{k+1} = x_k + s_k}.$$

Newton Solves the Linearized Equation

Set $F(x) = g(x) - y$. Starting from x_k , linearize:

$$F(x_k + s) \approx F(x_k) + J_g(x_k)s.$$

Impose $F(x_k) + J_g(x_k)s_k = 0$:

$$\boxed{J_g(x_k)s_k = y - g(x_k)}, \quad \boxed{x_{k+1} = x_k + s_k}.$$

Stop when, for chosen tolerances,

$$\|g(x_k) - y\| \leq \varepsilon_F \quad \text{or} \quad \|s_k\| \leq \varepsilon_x(1 + \|x_k\|).$$

Numerical practice

Solve for s_k ; never form $J_g(x_k)^{-1}$ explicitly. Newton is fast near a solution but may fail from a poor initial guess or a nearly singular Jacobian.

In One Dimension: Bisection or Newton?

For continuous $F : [a, b] \rightarrow \mathbb{R}$ with $F(a)F(b) < 0$, bisection repeats:

- 1 compute the midpoint $m = (a + b)/2$;
- 2 if $F(a)F(m) \leq 0$, replace b by m ;
- 3 otherwise, replace a by m ;
- 4 stop when the interval or residual is sufficiently small.

After k steps,

$$|x_k - x^*| \leq \frac{b - a}{2^{k+1}}.$$

Method	Main advantage	Main drawback
Bisection	guaranteed, derivative-free	linear and relatively slow
Newton	quadratic convergence locally	needs F' and a good initial guess

Common strategy

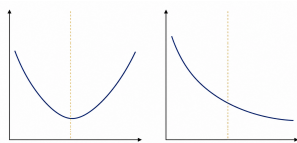
Bracket first with bisection, then switch to Newton near the root.

Implied Volatility Is an Implicit Function

For a call with spot S , strike K , maturity T , and constant rate r ,

$$C_{BS}(S, K, T, r, \sigma) = S\Phi(d_1) - Ke^{-rT}\Phi(d_2),$$

$$d_1 = \frac{\log(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$



Smile and skew across strikes.

Implied Volatility Is an Implicit Function

For a call with spot S , strike K , maturity T , and constant rate r ,

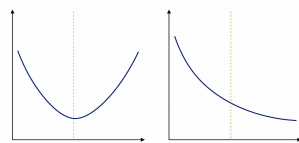
$$C_{BS}(S, K, T, r, \sigma) = S\Phi(d_1) - Ke^{-rT}\Phi(d_2),$$

$$d_1 = \frac{\log(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

Define

$$F(K, T, \sigma) = C_{BS}(S, K, T, r, \sigma) - C^{\text{mkt}}(K, T).$$

Implied volatility solves $F(K, T, \sigma_{\text{imp}}) = 0$.



Smile and skew across strikes.

Implicit Function Theorem

Since $\partial_{\sigma}F = \text{Vega} > 0$ for $T > 0$, the theorem defines a locally unique differentiable function $(K, T) \mapsto \sigma_{\text{imp}}(K, T)$.

5 Second Derivatives and Hessians

Hessian Matrix

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the Hessian is the matrix of second derivatives:

$$H_f(x) = \begin{pmatrix} \partial_{x_1 x_1}^2 f(x) & \cdots & \partial_{x_1 x_n}^2 f(x) \\ \vdots & \ddots & \vdots \\ \partial_{x_n x_1}^2 f(x) & \cdots & \partial_{x_n x_n}^2 f(x) \end{pmatrix}.$$

Hessian Matrix

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the Hessian is the matrix of second derivatives:

$$H_f(x) = \begin{pmatrix} \partial_{x_1 x_1}^2 f(x) & \cdots & \partial_{x_1 x_n}^2 f(x) \\ \vdots & \ddots & \vdots \\ \partial_{x_n x_1}^2 f(x) & \cdots & \partial_{x_n x_n}^2 f(x) \end{pmatrix}.$$

Schwarz theorem

If f is sufficiently regular, mixed partial derivatives commute:

$$\partial_{x_i x_j}^2 f = \partial_{x_j x_i}^2 f.$$

Then H_f is symmetric.

Hessian, Convexity, and Critical Points

Convexity criterion

If f is twice continuously differentiable on a convex set and

$$H_f(x) \succeq 0 \quad \text{for every } x,$$

then f is convex.

Hessian, Convexity, and Critical Points

Convexity criterion

If f is twice continuously differentiable on a convex set and

$$H_f(x) \succeq 0 \quad \text{for every } x,$$

then f is convex.

Criticality condition and second-order classification

The point x^* is critical when

$$\nabla f(x^*) = 0.$$

At such a point:

- $H_f(x^*) \succ 0$ implies a strict local minimum;
- $H_f(x^*) \prec 0$ implies a strict local maximum;
- an indefinite Hessian implies a saddle point.

Hessian Example

Let

$$f(x, y) = x^4 + 2x^2y^2 + y^4 = (x^2 + y^2)^2.$$

Compute

$$\nabla f(x, y) = \underline{\hspace{2cm}}, \quad H_f(x, y) = \underline{\hspace{2cm}}.$$

Hessian Example

Let

$$f(x, y) = x^4 + 2x^2y^2 + y^4 = (x^2 + y^2)^2.$$

Compute

$$\nabla f(x, y) = \underline{\hspace{2cm}}, \quad H_f(x, y) = \underline{\hspace{2cm}}.$$

$$\nabla f(x, y) = \begin{pmatrix} 4x(x^2 + y^2) \\ 4y(x^2 + y^2) \end{pmatrix}.$$

Hessian Example

Let

$$f(x, y) = x^4 + 2x^2y^2 + y^4 = (x^2 + y^2)^2.$$

Compute

$$\nabla f(x, y) = \underline{\hspace{2cm}}, \quad H_f(x, y) = \underline{\hspace{2cm}}.$$

$$\nabla f(x, y) = \begin{pmatrix} 4x(x^2 + y^2) \\ 4y(x^2 + y^2) \end{pmatrix}.$$

$$H_f(x, y) = \begin{pmatrix} 12x^2 + 4y^2 & 8xy \\ 8xy & 4x^2 + 12y^2 \end{pmatrix}.$$

Hessian Example

Let

$$f(x, y) = x^4 + 2x^2y^2 + y^4 = (x^2 + y^2)^2.$$

Compute

$$\nabla f(x, y) = \underline{\hspace{2cm}}, \quad H_f(x, y) = \underline{\hspace{2cm}}.$$

$$\nabla f(x, y) = \begin{pmatrix} 4x(x^2 + y^2) \\ 4y(x^2 + y^2) \end{pmatrix}.$$

$$H_f(x, y) = \begin{pmatrix} 12x^2 + 4y^2 & 8xy \\ 8xy & 4x^2 + 12y^2 \end{pmatrix}.$$

This Hessian is positive semidefinite, so f is convex.

Hands-On Example: Least Squares

For

$$L(x) = \frac{1}{2} \|Ax - b\|^2,$$

compute

$$\nabla L(x) = ?, \quad H_L(x) = ?.$$

Hands-On Example: Least Squares

For

$$L(x) = \frac{1}{2} \|Ax - b\|^2,$$

compute

$$\nabla L(x) = ?, \quad H_L(x) = ?.$$

Since

$$L(x) = \frac{1}{2} (Ax - b)^\top (Ax - b),$$

we obtain

$$\boxed{\nabla L(x) = A^\top (Ax - b)}, \quad \boxed{H_L(x) = A^\top A}.$$

Hands-On Example: Least Squares

For

$$L(x) = \frac{1}{2} \|Ax - b\|^2,$$

compute

$$\nabla L(x) = ?, \quad H_L(x) = ?.$$

Since

$$L(x) = \frac{1}{2} (Ax - b)^\top (Ax - b),$$

we obtain

$$\boxed{\nabla L(x) = A^\top (Ax - b)}, \quad \boxed{H_L(x) = A^\top A}.$$

With

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix},$$

find the least-squares minimizer.

Hands-On Example: Least Squares

For

$$L(x) = \frac{1}{2} \|Ax - b\|^2,$$

compute

$$\nabla L(x) = ?, \quad H_L(x) = ?.$$

Since

$$L(x) = \frac{1}{2} (Ax - b)^\top (Ax - b),$$

we obtain

$$\boxed{\nabla L(x) = A^\top (Ax - b)}, \quad \boxed{H_L(x) = A^\top A}.$$

With

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix},$$

find the least-squares minimizer. The normal equations are

$$\begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix} x = \begin{pmatrix} 5 \\ 6 \end{pmatrix}, \quad \boxed{x^* = \begin{pmatrix} 7/6 \\ 1/2 \end{pmatrix}}.$$

Full column rank makes $A^\top A$ positive definite, so the minimizer is unique.

6 Taylor Expansion

Taylor Expansion in One Dimension

Result – Taylor expansion at order k

If f is sufficiently smooth near x , then

$$f(x+h) = f(x) + f'(x)h + \frac{1}{2}f''(x)h^2 + \cdots + \frac{1}{k!}f^{(k)}(x)h^k + R_k(h).$$

Taylor Expansion in One Dimension

Result – Taylor expansion at order k

If f is sufficiently smooth near x , then

$$f(x+h) = f(x) + f'(x)h + \frac{1}{2}f''(x)h^2 + \cdots + \frac{1}{k!}f^{(k)}(x)h^k + R_k(h).$$

First and second order approximations

$$f(x+h) \approx f(x) + f'(x)h,$$

$$f(x+h) \approx f(x) + f'(x)h + \frac{1}{2}f''(x)h^2.$$

Taylor Examples

Around 0:

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots$$

$$\log(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$$

$$\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots$$

Taylor Examples

Around 0:

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots$$

$$\log(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$$

$$\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots$$

Use

Taylor expansions convert nonlinear problems into local polynomial approximations.

Taylor Remainder and Error Control

If f is $(k + 1)$ times continuously differentiable, then for some ξ between x and $x + h$,

$$f(x + h) = \sum_{j=0}^k \frac{f^{(j)}(x)}{j!} h^j + \frac{f^{(k+1)}(\xi)}{(k+1)!} h^{k+1}.$$

Taylor Remainder and Error Control

If f is $(k + 1)$ times continuously differentiable, then for some ξ between x and $x + h$,

$$f(x + h) = \sum_{j=0}^k \frac{f^{(j)}(x)}{j!} h^j + \frac{f^{(k+1)}(\xi)}{(k+1)!} h^{k+1}.$$

Use the quadratic Taylor polynomial to approximate $e^{0.1}$ and bound the remainder.

Taylor Remainder and Error Control

If f is $(k + 1)$ times continuously differentiable, then for some ξ between x and $x + h$,

$$f(x + h) = \sum_{j=0}^k \frac{f^{(j)}(x)}{j!} h^j + \frac{f^{(k+1)}(\xi)}{(k+1)!} h^{k+1}.$$

Use the quadratic Taylor polynomial to approximate $e^{0.1}$ and bound the remainder. The result is

$$1 + 0.1 + \frac{0.1^2}{2} = 1.105, \quad |R_2| \leq \frac{e^{0.1}}{6} (0.1)^3 < 1.85 \times 10^{-4}.$$

Message

Taylor's theorem provides both an approximation and a certified error bound.

Multivariate Taylor Expansion

Result – Second-order multivariate expansion

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and small $h \in \mathbb{R}^n$,

$$f(x + h) = f(x) + \nabla f(x)^\top h + \frac{1}{2} h^\top H_f(x) h + o(\|h\|^2).$$

Multivariate Taylor Expansion

Result – Second-order multivariate expansion

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and small $h \in \mathbb{R}^n$,

$$f(x + h) = f(x) + \nabla f(x)^\top h + \frac{1}{2} h^\top H_f(x) h + o(\|h\|^2).$$

Interpretation

- $\nabla f(x)^\top h$ is the local linear effect.
- $h^\top H_f(x) h$ is the local curvature effect.
- The Hessian controls second-order behavior.

Taylor and Local Optimization

Near a critical point x^* with $\nabla f(x^*) = 0$,

$$f(x^* + h) \approx f(x^*) + \frac{1}{2}h^\top H_f(x^*)h.$$

Taylor and Local Optimization

Near a critical point x^* with $\nabla f(x^*) = 0$,

$$f(x^* + h) \approx f(x^*) + \frac{1}{2}h^\top H_f(x^*)h.$$

- If $H_f(x^*)$ is positive definite, x^* is a local minimum.
- If $H_f(x^*)$ is negative definite, x^* is a local maximum.
- If $H_f(x^*)$ is indefinite, x^* is a saddle point.

Taylor and Local Optimization

Near a critical point x^* with $\nabla f(x^*) = 0$,

$$f(x^* + h) \approx f(x^*) + \frac{1}{2}h^\top H_f(x^*)h.$$

- If $H_f(x^*)$ is positive definite, x^* is a local minimum.
- If $H_f(x^*)$ is negative definite, x^* is a local maximum.
- If $H_f(x^*)$ is indefinite, x^* is a saddle point.

Bridge to Session 3

Optimization is built on gradients, Hessians, and Taylor approximations.

Section Roadmap

7 Integration

Integrals and the Fundamental Theorem

The integral $\int_a^b f(x) dx$ measures accumulated quantity over an interval.

Integrals and the Fundamental Theorem

The integral $\int_a^b f(x) dx$ measures accumulated quantity over an interval.

Result – Fundamental theorem of calculus

If $F'(x) = f(x)$, then

$$\int_a^b f(x) dx = F(b) - F(a).$$

Integrals and the Fundamental Theorem

The integral $\int_a^b f(x) dx$ measures accumulated quantity over an interval.

Result – Fundamental theorem of calculus

If $F'(x) = f(x)$, then

$$\int_a^b f(x) dx = F(b) - F(a).$$

Corollary – Differentiating the upper bound

If f is continuous and

$$G(x) = \int_a^x f(t) dt, \quad G'(x) = \frac{d}{dx} \int_a^x f(t) dt = f(x).$$

Example: Integrating Exponential Growth

Example

Compute $\int_0^T e^{rt} dt$ for $r \in \mathbb{R}$.

Example: Integrating Exponential Growth

Example

Compute $\int_0^T e^{rt} dt$ for $r \in \mathbb{R}$.

$$\int_0^T e^{rt} dt = \begin{cases} \frac{e^{rT} - 1}{r}, & r \neq 0, \\ T, & r = 0. \end{cases}$$

Example: Integrating Exponential Growth

Example

Compute $\int_0^T e^{rt} dt$ for $r \in \mathbb{R}$.

$$\int_0^T e^{rt} dt = \begin{cases} \frac{e^{rT} - 1}{r}, & r \neq 0, \\ T, & r = 0. \end{cases}$$

Why the case $r = 0$ is consistent

The limit of $(e^{rT} - 1)/r$ as $r \rightarrow 0$ equals T .

Change of Variables

Result – One-dimensional change of variables

If $u = \phi(x)$ and ϕ is differentiable, then

$$\int_a^b f(\phi(x))\phi'(x) dx = \int_{\phi(a)}^{\phi(b)} f(u) du.$$

Change of Variables

Result – One-dimensional change of variables

If $u = \phi(x)$ and ϕ is differentiable, then

$$\int_a^b f(\phi(x))\phi'(x) dx = \int_{\phi(a)}^{\phi(b)} f(u) du.$$

Example

$$\int_0^1 \frac{1}{\sqrt{1-x^2}} dx.$$

Choose a substitution and evaluate the integral. **Hint:** set $x = \sin u$ so that $1 - x^2 = \cos^2 u$.

Change of Variables

Result – One-dimensional change of variables

If $u = \phi(x)$ and ϕ is differentiable, then

$$\int_a^b f(\phi(x))\phi'(x) dx = \int_{\phi(a)}^{\phi(b)} f(u) du.$$

Example

$$\int_0^1 \frac{1}{\sqrt{1-x^2}} dx.$$

Choose a substitution and evaluate the integral. **Hint:** set $x = \sin u$ so that $1 - x^2 = \cos^2 u$. Use $x = \sin u$, so $dx = \cos u du$ and $u : 0 \rightarrow \pi/2$:

$$\int_0^{\pi/2} 1 du = \frac{\pi}{2}.$$

Change of Variables in $\mathbb{R}^n$

Let $\Phi : U \rightarrow V$ be a C^1 diffeomorphism between open subsets of $\mathbb{R}^n$. Then

Result – Multidimensional change of variables

$$\int_V f(y) \, dy = \int_U f(\Phi(x)) |\det J_\Phi(x)| \, dx.$$

More generally, for $D \subset U$,

$$\int_{\Phi(D)} f(y) \, dy = \int_D f(\Phi(x)) |\det J_\Phi(x)| \, dx.$$

Change of Variables in $\mathbb{R}^n$

Let $\Phi : U \rightarrow V$ be a C^1 diffeomorphism between open subsets of $\mathbb{R}^n$. Then

Result – Multidimensional change of variables

$$\int_V f(y) \, dy = \int_U f(\Phi(x)) |\det J_\Phi(x)| \, dx.$$

More generally, for $D \subset U$,

$$\int_{\Phi(D)} f(y) \, dy = \int_D f(\Phi(x)) |\det J_\Phi(x)| \, dx.$$

$|\det J_\Phi(x)|$ is the local factor by which Φ scales n -dimensional volume.

Polar Coordinates

For $\Phi(r, \theta) = (r \cos \theta, r \sin \theta)$,

$$|\det J_{\Phi}(r, \theta)| = r.$$

Evaluate the following integral over the disk of radius R :

$$\iint_{x^2+y^2 \leq R^2} (x^2 + y^2) \, dx \, dy.$$

Polar Coordinates

For $\Phi(r, \theta) = (r \cos \theta, r \sin \theta)$,

$$|\det J_{\Phi}(r, \theta)| = r.$$

Evaluate the following integral over the disk of radius R :

$$\iint_{x^2+y^2 \leq R^2} (x^2 + y^2) \, dx \, dy.$$

Using polar coordinates,

$$\iint_{x^2+y^2 \leq R^2} (x^2 + y^2) \, dx \, dy = \int_0^{2\pi} \int_0^R r^3 \, dr \, d\theta = \boxed{\frac{\pi R^4}{2}}.$$

Integration by Parts

Result – Integration by parts

For continuously differentiable functions u and v ,

$$\int_a^b u(x)v'(x) \, dx = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x) \, dx.$$

Integration by Parts

Result – Integration by parts

For continuously differentiable functions u and v ,

$$\int_a^b u(x)v'(x) \, dx = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x) \, dx.$$

Example

Evaluate $\int_0^T te^{-at} \, dt$ by parts, for $a > 0$.

Integration by Parts

Result – Integration by parts

For continuously differentiable functions u and v ,

$$\int_a^b u(x)v'(x) \, dx = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x) \, dx.$$

Example

Evaluate $\int_0^T te^{-at} \, dt$ by parts, for $a > 0$. Choose the factor to differentiate and the factor to integrate:

$$u(t) = t, \quad du = dt, \quad dv = e^{-at} dt, \quad v(t) = -\frac{1}{a}e^{-at}.$$

Integration by Parts

Result – Integration by parts

For continuously differentiable functions u and v ,

$$\int_a^b u(x)v'(x) dx = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x) dx.$$

Example

Evaluate $\int_0^T te^{-at} dt$ by parts, for $a > 0$. Choose the factor to differentiate and the factor to integrate:

$$u(t) = t, \quad du = dt, \quad dv = e^{-at} dt, \quad v(t) = -\frac{1}{a}e^{-at}.$$

Therefore

$$\int_0^T te^{-at} dt = \left[-\frac{t}{a}e^{-at}\right]_0^T + \frac{1}{a} \int_0^T e^{-at} dt = \boxed{\frac{1 - e^{-aT}(1 + aT)}{a^2}}.$$

Result – Leibniz rule

Let

$$F(\theta) = \int_{a(\theta)}^{b(\theta)} f(x, \theta) dx.$$

Leibniz Rule

Result – Leibniz rule

Let

$$F(\theta) = \int_{a(\theta)}^{b(\theta)} f(x, \theta) dx.$$

Then

$$F'(\theta) = f(b(\theta), \theta)b'(\theta) - f(a(\theta), \theta)a'(\theta) + \int_{a(\theta)}^{b(\theta)} \partial_{\theta} f(x, \theta) dx.$$

Leibniz Rule

Result – Leibniz rule

Let

$$F(\theta) = \int_{a(\theta)}^{b(\theta)} f(x, \theta) dx.$$

Then

$$F'(\theta) = f(b(\theta), \theta)b'(\theta) - f(a(\theta), \theta)a'(\theta) + \int_{a(\theta)}^{b(\theta)} \partial_{\theta} f(x, \theta) dx.$$

Finance link

Greeks often come from differentiating prices written as expectations or integrals.

Sensitivity of an Expectation

Suppose

$$P(\theta) = \mathbb{E}[g(X_\theta)] = \int_{\mathbb{R}} g(x)p(x, \theta) \, dx.$$

Sensitivity of an Expectation

Suppose

$$P(\theta) = \mathbb{E}[g(X_\theta)] = \int_{\mathbb{R}} g(x)p(x, \theta) \, dx.$$

When the interchange is justified,

$$\frac{\partial P}{\partial \theta} = \int_{\mathbb{R}} g(x) \partial_\theta p(x, \theta) \, dx.$$

Sensitivity of an Expectation

Suppose

$$P(\theta) = \mathbb{E}[g(X_\theta)] = \int_{\mathbb{R}} g(x)p(x, \theta) dx.$$

When the interchange is justified,

$$\frac{\partial P}{\partial \theta} = \int_{\mathbb{R}} g(x)\partial_\theta p(x, \theta) dx.$$

Interpretation

Differentiating under the integral sign turns pricing sensitivities into calculus problems.

- 8 Ordinary Differential Equations
 - Theoretical Results
 - General Scalar Calculations
 - Linear Systems in Several Dimensions
 - Second-Order Equations
 - Numerical Methods

Ordinary Differential Equations

An ordinary differential equation describes the time evolution of an unknown function.

$$y'(t) = f(t, y(t)), \quad y(0) = y_0.$$

Ordinary Differential Equations

An ordinary differential equation describes the time evolution of an unknown function.

$$y'(t) = f(t, y(t)), \quad y(0) = y_0.$$

- The derivative gives the instantaneous direction of motion.
- The initial condition selects one trajectory.
- Financial state variables often evolve through differential equations.

Existence and Uniqueness of an ODE

Result – Picard–Lindelöf

For

$$y'(t) = f(t, y(t)), \quad y(t_0) = y_0,$$

Picard–Lindelöf says:

- continuity of f gives local existence;
- local Lipschitz continuity in y gives local uniqueness.

How to check the hypothesis

Near every (t_0, y_0) , there are a neighborhood and $L < \infty$ such that

$$|f(t, y_1) - f(t, y_2)| \leq L|y_1 - y_2|$$

for points in that neighborhood. A continuous $\partial_y f$ is sufficient.

Existence and Uniqueness of an ODE

Result – Picard–Lindelöf

For

$$y'(t) = f(t, y(t)), \quad y(t_0) = y_0,$$

Picard–Lindelöf says:

- continuity of f gives local existence;
- local Lipschitz continuity in y gives local uniqueness.

How to check the hypothesis

Near every (t_0, y_0) , there are a neighborhood and $L < \infty$ such that

$$|f(t, y_1) - f(t, y_2)| \leq L|y_1 - y_2|$$

for points in that neighborhood. A continuous $\partial_y f$ is sufficient.

Without the Lipschitz condition, uniqueness may fail.

The Fundamental Solution Propagates Initial Data

Result – Homogeneous scalar linear ODE

Start with the homogeneous equation

$$y'(t) = a(t)y(t), \quad y(0) = y_0.$$

Its fundamental solution is

$$\Phi(t, 0) = \exp \left(\int_0^t a(u) \, du \right).$$

The Fundamental Solution Propagates Initial Data

Result – Homogeneous scalar linear ODE

Start with the homogeneous equation

$$y'(t) = a(t)y(t), \quad y(0) = y_0.$$

Its fundamental solution is

$$\Phi(t, 0) = \exp \left(\int_0^t a(u) \, du \right).$$

It satisfies

$$\partial_t \Phi(t, 0) = a(t)\Phi(t, 0), \quad \Phi(0, 0) = 1.$$

Therefore

$$y(t) = \Phi(t, 0)y_0.$$

Variation of the Constant Adds the Forcing

Consider

$$y'(t) = a(t)y(t) + b(t), \quad y(0) = y_0.$$

Seek $y(t) = c(t)\Phi(t, 0)$. Then

$$y' = c'\Phi + c a(t)\Phi.$$

Variation of the Constant Adds the Forcing

Consider

$$y'(t) = a(t)y(t) + b(t), \quad y(0) = y_0.$$

Seek $y(t) = c(t)\Phi(t, 0)$. Then

$$y' = c'\Phi + c a(t)\Phi.$$

Substitution gives

$$c'(t)\Phi(t, 0) = b(t), \quad c(t) = y_0 + \int_0^t \Phi(s, 0)^{-1} b(s) \, ds.$$

Using $\Phi(t, 0)\Phi(s, 0)^{-1} = \Phi(t, s)$,

Result – Scalar variation-of-constants formula

$$y(t) = \Phi(t, 0)y_0 + \int_0^t \Phi(t, s)b(s) \, ds.$$

Linear ODE: Exponential Growth and Decay

Consider

$$y'(t) = ky(t), \quad y(0) = y_0.$$

Linear ODE: Exponential Growth and Decay

Consider

$$y'(t) = ky(t), \quad y(0) = y_0.$$

The solution is

$$y(t) = y_0 e^{kt}.$$

Linear ODE: Exponential Growth and Decay

Consider

$$y'(t) = ky(t), \quad y(0) = y_0.$$

The solution is

$$y(t) = y_0 e^{kt}.$$

- $k > 0$: exponential growth.
- $k < 0$: exponential decay.
- $k = 0$: constant state.

Worked Linear ODE Examples

For

$$y' - 2y = t, \quad y(0) = 1,$$

solve by variation of the constant.

Worked Linear ODE Examples

For

$$y' - 2y = t, \quad y(0) = 1,$$

solve by variation of the constant. Variation of the constant gives

$$y(t) = e^{2t} \left(1 + \int_0^t s e^{-2s} ds \right) = \boxed{\frac{5}{4} e^{2t} - \frac{1 + 2t}{4}}.$$

Worked Linear ODE Examples

For

$$y' - 2y = t, \quad y(0) = 1,$$

solve by variation of the constant. Variation of the constant gives

$$y(t) = e^{2t} \left(1 + \int_0^t s e^{-2s} ds \right) = \boxed{\frac{5}{4} e^{2t} - \frac{1 + 2t}{4}}.$$

For

$$y' + \frac{2}{1+t} y = 1 + t, \quad y(0) = 1,$$

solve with an integrating factor.

Worked Linear ODE Examples

For

$$y' - 2y = t, \quad y(0) = 1,$$

solve by variation of the constant. Variation of the constant gives

$$y(t) = e^{2t} \left(1 + \int_0^t s e^{-2s} ds \right) = \boxed{\frac{5}{4} e^{2t} - \frac{1 + 2t}{4}}.$$

For

$$y' + \frac{2}{1+t} y = 1 + t, \quad y(0) = 1,$$

solve with an integrating factor. The integrating factor is $(1+t)^2$, and

$$y(t) = \boxed{\frac{(1+t)^4 + 3}{4(1+t)^2}}.$$

Multidimensional Linear ODEs

A linear system has the form

$$x'(t) = A(t)x(t) + g(t), \quad x(t_0) = x_0,$$

with $x(t) \in \mathbb{R}^d$ and $A(t) \in \mathbb{R}^{d \times d}$.

Multidimensional Linear ODEs

A linear system has the form

$$x'(t) = A(t)x(t) + g(t), \quad x(t_0) = x_0,$$

with $x(t) \in \mathbb{R}^d$ and $A(t) \in \mathbb{R}^{d \times d}$. If A and g are continuous, the system has a unique solution on every compact interval where they are defined.

Result – Homogeneous constant-coefficient system

When A is constant in time, $x' = Ax$ has solution

$$x(t) = e^{tA}x_0, \quad e^{tA} = I + tA + \frac{t^2 A^2}{2!} + \dots$$

For time-dependent $A(t)$, one cannot generally replace the solution by $e^{\int_0^t A(s) ds}$ because matrices at different times may not commute.

Diagonalization Computes the Matrix Exponential

Result – Matrix exponential by diagonalization

If $A = PDP^{-1}$ with $D = \text{diag}(\lambda_1, \dots, \lambda_d)$, then

$$\boxed{e^{tA} = Pe^{tD}P^{-1}}, \quad e^{tD} = \text{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_d t}).$$

Diagonalization Computes the Matrix Exponential

Result – Matrix exponential by diagonalization

If $A = PDP^{-1}$ with $D = \text{diag}(\lambda_1, \dots, \lambda_d)$, then

$$\boxed{e^{tA} = Pe^{tD}P^{-1}}, \quad e^{tD} = \text{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_d t}).$$

Each eigenvalue creates one dynamical mode:

- $\text{Re}(\lambda) < 0$: decay;
- $\text{Re}(\lambda) > 0$: growth;
- $\text{Im}(\lambda) \neq 0$: oscillation.

Stability criterion

$x = 0$ is asymptotically stable when every eigenvalue of A has strictly negative real part.

Result – Multidimensional variation of constants

For $x'(t) = Ax(t) + g(t)$ with $x(0) = x_0$,

$$x(t) = e^{tA}x_0 + \int_0^t e^{(t-s)A}g(s) \, ds.$$

Variation of Constants in $\mathbb{R}^d$

Result – Multidimensional variation of constants

For $x'(t) = Ax(t) + g(t)$ with $x(0) = x_0$,

$$x(t) = e^{tA}x_0 + \int_0^t e^{(t-s)A}g(s) \, ds.$$

For constant forcing $g(t) = b$ and invertible A , let $x^* = -A^{-1}b$. Then

$$x(t) = x^* + e^{tA}(x_0 - x^*).$$

Interpretation

The exponential propagates the initial displacement; the integral accumulates the forcing.

Worked Two-Dimensional Linear System

Consider

$$x' = \begin{pmatrix} -1 & 1 \\ 0 & -2 \end{pmatrix} x, \quad x(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Find the solution by decomposing the initial condition into eigenvectors.

Worked Two-Dimensional Linear System

Consider

$$x' = \begin{pmatrix} -1 & 1 \\ 0 & -2 \end{pmatrix} x, \quad x(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Find the solution by decomposing the initial condition into eigenvectors. The eigenpairs are

$$\lambda_1 = -1, \quad v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \lambda_2 = -2, \quad v_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

Since $x_0 = v_1 + v_2$,

$$x(t) = \begin{pmatrix} e^{-t} - e^{-2t} \\ e^{-2t} \end{pmatrix}.$$

Both modes decay; e^{-t} dominates in the long run.

Second-Order Linear ODEs

For

$$y'' + ay' + by = 0,$$

the trial solution e^{rt} gives the characteristic equation

$$r^2 + ar + b = 0.$$

Second-Order Linear ODEs

For

$$y'' + ay' + by = 0,$$

the trial solution e^{rt} gives the characteristic equation

$$r^2 + ar + b = 0.$$

Distinct roots r_1, r_2 give $y = C_1 e^{r_1 t} + C_2 e^{r_2 t}$; complex roots give damped oscillations.

For

$$y'' + 3y' + 2y = 0, \quad y(0) = 1, \quad y'(0) = 0,$$

find the solution satisfying the two initial conditions.

Second-Order Linear ODEs

For

$$y'' + ay' + by = 0,$$

the trial solution e^{rt} gives the characteristic equation

$$r^2 + ar + b = 0.$$

Distinct roots r_1, r_2 give $y = C_1 e^{r_1 t} + C_2 e^{r_2 t}$; complex roots give damped oscillations.

For

$$y'' + 3y' + 2y = 0, \quad y(0) = 1, \quad y'(0) = 0,$$

find the solution satisfying the two initial conditions. the roots are $-1, -2$ and

$$y(t) = 2e^{-t} - e^{-2t}.$$

A Second-Order ODE Is a First-Order System

Set $x_1 = y$ and $x_2 = y'$. Then

$$y'' + ay' + by = f(t)$$

is equivalent to

$$\begin{pmatrix} x_1' \\ x_2' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -b & -a \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ f(t) \end{pmatrix}.$$

A Second-Order ODE Is a First-Order System

Set $x_1 = y$ and $x_2 = y'$. Then

$$y'' + ay' + by = f(t)$$

is equivalent to

$$\begin{pmatrix} x_1' \\ x_2' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -b & -a \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ f(t) \end{pmatrix}.$$

The characteristic roots of $r^2 + ar + b = 0$ are exactly the eigenvalues of the companion matrix.

Unified viewpoint

Coupled equations and higher-order scalar equations are handled with the same matrix tools.

Explicit and Implicit Euler

For $y' = f(t, y)$ and $t_{n+1} = t_n + h$:

$$\boxed{y_{n+1} = y_n + hf(t_n, y_n)} \quad (\text{explicit}),$$

$$\boxed{y_{n+1} = y_n + hf(t_{n+1}, y_{n+1})} \quad (\text{implicit}).$$

Explicit and Implicit Euler

For $y' = f(t, y)$ and $t_{n+1} = t_n + h$:

$$y_{n+1} = y_n + hf(t_n, y_n) \quad (\text{explicit}),$$

$$y_{n+1} = y_n + hf(t_{n+1}, y_{n+1}) \quad (\text{implicit}).$$

Result – First-order convergence

If f is continuous in t , Lipschitz in y , and sufficiently smooth along the exact solution, then on $[0, T]$

$$\max_{0 \leq n \leq N} |y(t_n) - y_n| \leq C_T h.$$

Both schemes are therefore first-order convergent. For implicit Euler, $hL < 1$ is a simple sufficient condition for the step equation to have a unique solution when f is L -Lipschitz in y .

Euler Schemes for Linear Systems

For $x' = Ax + g(t)$, explicit Euler gives

$$x_{n+1} = (I + hA)x_n + hg(t_n).$$

Euler Schemes for Linear Systems

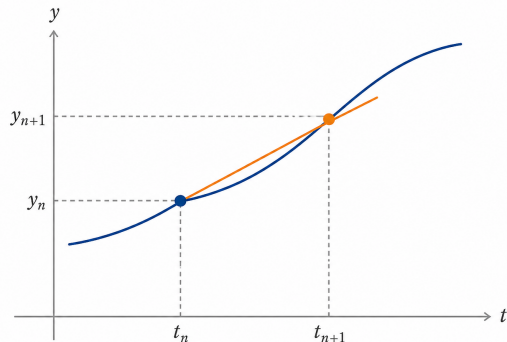
For $x' = Ax + g(t)$, explicit Euler gives

$$x_{n+1} = (I + hA)x_n + hg(t_n).$$

Implicit Euler gives the linear system

$$(I - hA)x_{n+1} = x_n + hg(t_{n+1}).$$

If A is constant, implicit Euler is well defined whenever $I - hA$ is invertible.



Explicit Euler follows the tangent over one time step.

Euler Stability on $y' = \lambda y$

Explicit Euler gives the recurrence

$$y_{n+1} = (1 + h\lambda)y_n.$$

For real $\lambda < 0$, decay requires

$$|1 + h\lambda| < 1 \quad \Longleftrightarrow \quad 0 < h < -\frac{2}{\lambda}.$$

Euler Stability on $y' = \lambda y$

Explicit Euler gives the recurrence

$$y_{n+1} = (1 + h\lambda)y_n.$$

For real $\lambda < 0$, decay requires

$$|1 + h\lambda| < 1 \iff 0 < h < -\frac{2}{\lambda}.$$

Implicit Euler gives

$$y_{n+1} = \frac{1}{1 - h\lambda} y_n,$$

and the multiplier has modulus below one for every $h > 0$ when $\lambda < 0$. Thus, for real $\lambda < 0$,

Result – Stability domains on the negative real axis

explicit: $0 < h < -2/\lambda$, implicit: every $h > 0$.

9 Financial Models

Risk-Free Asset

A risk-free bank account B_t with a prescribed short-rate curve $r(t)$ satisfies

$$\frac{dB_t}{dt} = r(t)B_t, \quad B_0 > 0.$$

Risk-Free Asset

A risk-free bank account B_t with a prescribed short-rate curve $r(t)$ satisfies

$$\frac{dB_t}{dt} = r(t)B_t, \quad B_0 > 0.$$

The solution is

$$B_t = B_0 \exp \left(\int_0^t r(s) ds \right).$$

Risk-Free Asset

A risk-free bank account B_t with a prescribed short-rate curve $r(t)$ satisfies

$$\frac{dB_t}{dt} = r(t)B_t, \quad B_0 > 0.$$

The solution is

$$B_t = B_0 \exp \left(\int_0^t r(s) ds \right).$$

For a constant rate r :

$$B_t = B_0 e^{rt}.$$

Why is this an ODE?

Here $r(t)$ is a fixed input known as a function of time. If the short rate is random, its dynamics require an SDE and pricing generally involves conditional expectations.

Continuous Discounting

If B_t is the risk-free account, the discount factor from 0 to T is

$$D(0, T) = \frac{B_0}{B_T} = \exp \left(- \int_0^T r(s) \, ds \right).$$

Continuous Discounting

If B_t is the risk-free account, the discount factor from 0 to T is

$$D(0, T) = \frac{B_0}{B_T} = \exp \left(- \int_0^T r(s) \, ds \right).$$

For constant r ,

$$D(0, T) = e^{-rT}.$$

Continuous Discounting

If B_t is the risk-free account, the discount factor from 0 to T is

$$D(0, T) = \frac{B_0}{B_T} = \exp \left(- \int_0^T r(s) \, ds \right).$$

For constant r ,

$$D(0, T) = e^{-rT}.$$

Pricing intuition

The present value of a deterministic payoff X_T at time T is

$$X_0 = D(0, T)X_T.$$

Mean-Reverting ODE

The deterministic mean-reversion equation is

$$r'(t) = a(b - r(t)), \quad a > 0.$$

Mean-Reverting ODE

The deterministic mean-reversion equation is

$$r'(t) = a(b - r(t)), \quad a > 0.$$

- b is the long-run level.
- a is the speed of mean reversion.
- If $r(t) > b$, the derivative is negative.
- If $r(t) < b$, the derivative is positive.

Mean-Reverting ODE

The deterministic mean-reversion equation is

$$r'(t) = a(b - r(t)), \quad a > 0.$$

- b is the long-run level.
- a is the speed of mean reversion.
- If $r(t) > b$, the derivative is negative.
- If $r(t) < b$, the derivative is positive.

Solution:

$$r(t) = b + (r_0 - b)e^{-at}.$$

Vasicek Short-Rate Model

The stochastic Vasicek model adds Brownian noise:

$$dr_t = a(b - r_t) dt + \sigma dW_t.$$

Vasicek Short-Rate Model

The stochastic Vasicek model adds Brownian noise:

$$dr_t = a(b - r_t) dt + \sigma dW_t.$$

The conditional mean solves the deterministic mean-reversion ODE:

$$\frac{d}{dt} \mathbb{E}[r_t | r_s] = a(b - \mathbb{E}[r_t | r_s]).$$

Vasicek Short-Rate Model

The stochastic Vasicek model adds Brownian noise:

$$dr_t = a(b - r_t) dt + \sigma dW_t.$$

The conditional mean solves the deterministic mean-reversion ODE:

$$\frac{d}{dt} \mathbb{E}[r_t | r_s] = a(b - \mathbb{E}[r_t | r_s]).$$

$$\mathbb{E}[r_t | r_s] = b + (r_s - b)e^{-a(t-s)}.$$

Discounting Along the Vasicek Mean Path

If we use the conditional mean path $m(t) = b + (r_0 - b)e^{-at}$ as the short-rate curve, then

$$\int_0^T m(t) dt = bT + \frac{r_0 - b}{a}(1 - e^{-aT}).$$

Discounting Along the Vasicek Mean Path

If we use the conditional mean path $m(t) = b + (r_0 - b)e^{-at}$ as the short-rate curve, then

$$\int_0^T m(t) dt = bT + \frac{r_0 - b}{a}(1 - e^{-aT}).$$

The discount factor is

$$D(0, T) = \exp \left(-bT - \frac{r_0 - b}{a}(1 - e^{-aT}) \right).$$

Discounting Along the Vasicek Mean Path

If we use the conditional mean path $m(t) = b + (r_0 - b)e^{-at}$ as the short-rate curve, then

$$\int_0^T m(t) dt = bT + \frac{r_0 - b}{a}(1 - e^{-aT}).$$

The discount factor is

$$D(0, T) = \exp\left(-bT - \frac{r_0 - b}{a}(1 - e^{-aT})\right).$$

Takeaway

This computes discounting along the mean path. It is not the stochastic Vasicek bond price, which requires an expectation of the random discount factor.